

MYSQL Backup IN ENTERPRISE EDITION

1. Prepare backup directories

```
sudo mkdir -p /backup/mysql/{full,incremental,binlog}
sudo chown -R mysql:mysql /backup/mysql
sudo chmod 700 /backup/mysql
```

Check MEB:

```
mysqlbackup --version
```

2. Create a full backup

I recommend using a dedicated backup user rather than MySQL root.

Example:

```
CREATE USER 'mysqlbackup'@'localhost'
IDENTIFIED BY 'STRONG_PASSWORD';
```

```
GRANT BACKUP_ADMIN ON *.* TO 'mysqlbackup'@'localhost';
```

Then perform a full backup :

```
mysqlbackup \
--user=mysqlbackup \
--password \
--backup-dir=/backup/mysql/full \
backup-and-apply-log
```

Note : This is hot physical backup mysql remain online .

Check it : `ls -lah /backup/mysql/full`

Validate the backup :

```
mysqlbackup --backup-dir=/backup/mysql/full validate
```

3. Create an incremental backup

Suppose your full backup is:

`/backup/mysql/full`

Create first incremental backup :

```
mysqlbackup \
--user=mysqlbackup \
--password \
--backup-dir=/backup/mysql/incremental/inc-01 \
--incremental \
--incremental-base=dir:/backup/mysql/full \
backup
```

Then the next incremental can be based on the previous incremental:

```
mysqlbackup \  
--user=mysqlbackup \  
--password \  
--backup-dir=/backup/mysql/incremental/inc-02 \  
--incremental \  
--incremental-base=dir:/backup/mysql/incremental/inc-01 \  
backup
```

Menas:

FULL

```
|— INC-01  
|— INC-02  
|— INC-03  
└— INC-04
```

4. Enable binary logging

For PITR, make sure binary logging is enabled.

Check:

```
SHOW VARIABLES LIKE 'log_bin';
```

Must Be : ON

log_bin = ON

[mysqld]

server_id=1001

log_bin=/var/lib/mysql/binlog/mysql-bin (by default it created on instllation)

binlog_format=ROW

binlog_expire_logs_seconds=604800

Create a directory :

```
sudo mkdir -p /var/lib/mysql/binlog
```

```
sudo chown mysql:mysql /var/lib/mysql/binlog
```

Then restart

```
sudo systemctl restart mysqld
```

Verify :

```
SHOW VARIABLES LIKE 'log_bin';
```

```
SHOW BINARY LOGS;
```

Note : Do not delete binary logs before you're sure they are no longer required for your recovery window.

5. Capture the binary logs

For PITR, you need to preserve binary logs somewhere separate from the database server.

For example, you might periodically copy them to:

```
/backup/mysql/binlog/
```

```
SHOW BINARY LOG STATUS;
```

or

```
SHOW BINARY LOGS;
```

6. Restore a full backup

Suppose the server has failed and you've installed MySQL 8.4 on a new Oracle Linux server.
Stop MySQL:

```
sudo systemctl stop mysqld
```

Note : Make sure the datadir is empty/new:

```
sudo ls -lah /var/lib/mysql
```

Then restore the full backup :

```
mysqlbackup \  
--backup-dir=/backup/mysql/full \  
--datadir=/var/lib/mysql \  
copy-back
```

Or use this to apply and restore simultaneously

```
mysqlbackup \  
--backup-dir=/backup/mysql/full \  
--datadir=/var/lib/mysql \  
copy-back-and-apply-log
```

Then fix the ownership

```
sudo chown -R mysql:mysql /var/lib/mysql
```

Start mysql :

```
sudo systemctl start mysqld
```

Check the status :

```
sudo systemctl status mysqld
```

7. Restore full + incremental backups

This is where the sequence matters.

Suppose you have:

```
FULL  
INC-01  
INC-02  
INC-03
```

You restore the full backup first.

Then apply the incremental backups in order.

For example:

```
mysqlbackup \  
--backup-dir=/backup/mysql/full \  
apply-log
```

Here : apply-log meaning is

It means roughly:

"Prepare the physical backup using the redo information associated with the backup."

Then apply the incrementals according to the MEB incremental-recovery procedure.

The important concept is:

```
FULL
↓
INC-01
↓
INC-02
↓
INC-03
↓
READY FOR DATABASE START
```

Note : Don't randomly apply INC-03 without the preceding incremental chain.

8. PITR — Point-in-Time Recovery

This is the most important part.

Scenario :

Monday 01:00

```
|
└─ Full backup
```

```
Tuesday |
└─ Incremental
```

Wednesday 14:00

```
|
└─ Database changes
```

```
|
└─ 14:30
```

```
|
└─ 14:35
```

```
|
└─ 14:37 <-- accidental DROP TABLE
```

We want to recover up to Wednesday 14:36:59

Then we apply

FULL BACKUP

↓

INCREMENTALS

↓

BINARY LOGS

↓

STOP BEFORE 14:37

↓

RECOVERED DATABASE

9. Find the correct binary logs

First determine which binlog contains the incident.

For example

```
mysqlbinlog \
--base64-output=DECODE-ROWS \
-vv \
```

mysql-bin.000125

10. Apply binary logs until a specific time

After restoring the physical backup, you can use mysqlbinlog to apply the binary logs.

For example:

```
mysqlbinlog \  
--stop-datetime="2026-08-26 14:37:01" \  
/backup/mysql/binlog/mysql-bin.000125 \  
| mysql -uroot -p
```

If multiple binlog are present then

```
mysqlbinlog \  
/backup/mysql/binlog/mysql-bin.000123 \  
/backup/mysql/binlog/mysql-bin.000124 \  
--stop-datetime="2026-08-26 14:37:01" \  
| mysql -uroot -p
```

Main is :

--stop-datetime

This allows you to stop replaying transactions at a particular time.

11. PITR using a position

Time-based recovery isn't always the best option.

You can also recover to a specific binary-log position:

```
mysqlbinlog \  
--stop-position=123456789 \  
/backup/mysql/binlog/mysql-bin.000125 \  
| mysql -uroot -p
```

This is often preferable when you've identified the exact transaction/event that caused the problem.

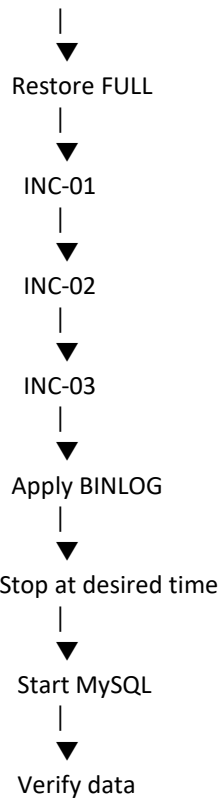
Complete recovery plan :

```
/backup/mysql/  
|  
|— full/  
|  
|— incremental/  
| |— inc-01/  
| |— inc-02/  
| |— inc-03/  
|  
|— binlog/  
| |— mysql-bin.000120  
| |— mysql-bin.000121  
| |— mysql-bin.000122  
| |— mysql-bin.000123
```

FAILURE



Install MySQL



Note :

- Sunday: Full MEB backup
- Monday–Saturday: Incremental MEB backups
- Continuously: Preserve binary logs
- Daily: Copy backups/binlogs off the database server
- Monthly: Perform a complete restore test
- Quarterly: Test PITR/DR

Restore flow :

```
mysqlbackup \  
--backup-dir=/backup/mysql/full \  
apply-log
```

```
mysqlbackup \  
--backup-dir=/backup/mysql/full \  
--incremental-backup-dir=/backup/mysql/incremental/inc-01 \  
apply-incremental-backup
```

```
mysqlbackup \  
--backup-dir=/backup/mysql/full \  
--incremental-backup-dir=/backup/mysql/incremental/inc-02 \  
apply-incremental-backup
```

```
mysqlbackup \  
--backup-dir=/backup/mysql/full \  
--incremental-backup-dir=/backup/mysql/incremental/inc-03 \  
apply-incremental-backup
```